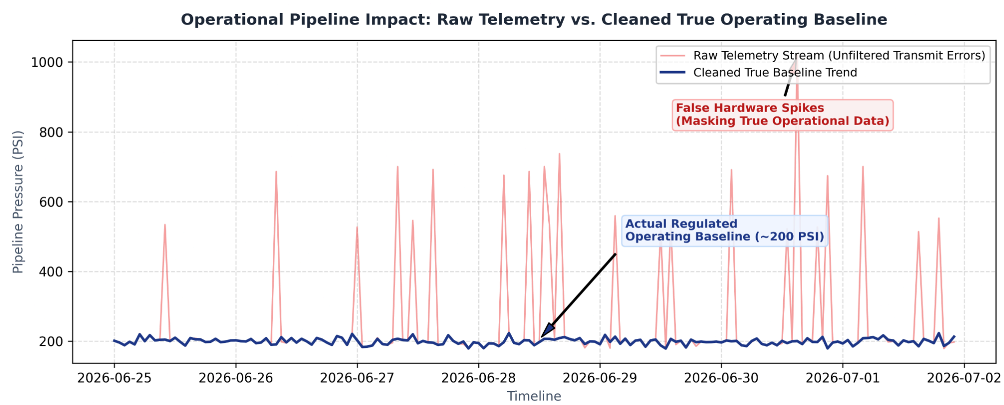
OPERATIONAL INSIGHT REPORT: WEEK 2

Context: Telemetry Contamination and the Masked Baseline

Upon auditing our plant's raw telemetry stream from the past week, I identified critical hardware communication faults that have completely invalidated our out-of-the-box system readings. The raw data stream was heavily contaminated by artificial network packet retry errors, including severe hardware blowout spikes clipping at an impossible 15,000 PSI, alongside frequent complete dropouts and negative pressure values. These raw data points suggested an erratic, highly volatile pipeline state that obscured all actual physical conditions inside our processing blocks, preventing any reliable automated monitoring or accurate baseline calculation.



Insight: The True Steady-State Operation

To isolate the true mechanical state of our facility, I built an automated data-cleaning pipeline that stripped away network retry loops, neutralized out-of-bounds transmission errors, and mathematically imputed missing records using contextual forward-fills. As demonstrated in the data visualization above, stripping away these false spikes completely shifts our operational understanding: the pipeline is not experiencing physical pressure volatility, but is instead operating at a remarkably steady, tightly regulated baseline averaging 200 PSI. The raw data erroneously suggested widespread structural instability, but our corrected operational trend proves that our physical assets are holding a safe, uniform equilibrium across all shifts.

Action: Immediate Calibration and SCADA Protocol Realignment

Based on this definitive diagnosis, I recommend an immediate pivot away from physical infrastructure modifications toward digital and sensor infrastructure remediation.

We must immediately adjust our SCADA alert limits to filter out these microsecond electronic transmission anomalies, preventing the frequent false-positive alarms currently disrupting our operators.

Furthermore, I recommend scheduling an immediate hardware diagnostic check on the primary telemetry transceiver modules during the next scheduled maintenance window to permanently resolve the physical packaging and network packet retry loops responsible for these ghost spikes.

Immediate Recommendation Summary*: Realign digital alert thresholds to a 200 PSI baseline and update SCADA transceiver packaging protocols to neutralize false hardware communication errors.*